

Style-Aware Adaptation for Language-Conditioned Humanoid Motion Generation

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Abstract

Language-conditioned humanoid motion generation provides an intuitive interface for controlling animated characters and humanoid robots. This project studies style-aware adaptation of MoConVQ, a physics-based motion generation framework with discrete motion representations. Since the proposal, we have reproduced the pretrained inference pipeline, verified motion tokenization and decoding, and generated baseline motions for ten generic and style-related prompts. We also implemented the data-curation and training-diagnostic entry points needed for a HumanML3D style subset. HumanML3D is not yet present in the local workspace, so we report the current stage as an honest diagnostic midterm: baseline inference is complete, style-subset tooling is ready, and fine-tuning is gated on preparing text-motion annotations and MoConVQ token targets.

1 Problem and Motivation

Given a natural-language prompt, the goal is to generate a 3D humanoid motion sequence that captures both the action semantics and the requested style. MoConVQ is a suitable robotics baseline because it combines discrete motion representations with physics-based motion control [3]. However, style-specific prompts such as slow deliberate motion, tai chi-like motion, or graceful dance may remain underspecified by a general text-to-motion model. Our project therefore focuses on parameter-efficient style adaptation under limited compute, while avoiding training the motion representation backbone from scratch.

2 Related Work

HumanML3D provides paired motion and text descriptions for language-conditioned motion generation and is the planned source for our style-focused subset [1]. MDM represents diffusion-based raw-motion generation [2], while T2M-GPT shows the effectiveness of discrete motion tokens for text-to-motion generation [4]. Our work differs from full model training: we keep MoConVQ’s motion representation and decoder fixed and adapt only the language-conditioned generation component when tokenized supervision is available.

3 Technical Approach

The baseline inference pipeline is:

Text prompt → T5 text encoder → language transformer
→ discrete motion code sequence → MoConVQ
decoder/controller → BVH humanoid motion

Figure 1: Project pipeline. The motion representation backbone and physics-based decoder are kept frozen; adaptation is limited to the text-conditioned code generator.

For data curation, we preserve the official HumanML3D train, validation, and test split files. Keyword filtering is applied independently inside each split using a lexicon with controlled, graceful, dance, energetic, and heavy/awkward style groups. This prevents leakage from validation or test captions into training. A 30-sample manual audit table is generated to distinguish true style-related captions from ambiguous keyword matches.

For adaptation, the intended objective is cross-entropy over ground-truth MoConVQ token sequences. The upstream MoConVQ release provides pretrained inference weights but not the original MoConGPT training code, so our current implementation includes a diagnostic training entry point that checks for style captions, token targets, and pretrained files before enabling optimization.

4 Preliminary Results

Table 1 summarizes current implementation status. Baseline reproduction is complete: tokenization and decoding both run locally, the default text-to-motion prompt runs, and ten project prompts generate BVH files under the MoConVQ output directory. The generated prompt set includes both generic actions and style-related prompts such as deliberate walking, graceful dancing, tai chi-like motion, energetic jumping, and tired heavy walking.

5 Remaining Work

The next milestone is to place HumanML3D annotations under `datasets/HumanML3D`, run the style-subset filter, manually audit sampled captions, and tokenize selected motions into MoConVQ code sequences. If tokenization succeeds, we will fine-tune the language-conditioned generator with fp16 and

Table 1: Implementation progress at midterm.

Component	Status	Evidence
Environment	Complete	CUDA PyTorch and local extensions verified
Baseline inference	Complete	Token/decode and 10 BVHs generated
Style subset tooling	Ready	Split-preserving filter and audit sampler
HumanML3D data	Pending	Expected under datasets/
Fine-tuning loop	Diagnostic	Waiting for MoConVQ token targets

Table 2: Style-subset status. Counts will be filled by the filtering script after HumanML3D is placed locally.

Split	Matched cap-tions	Matched motions
Train	pending	pending
Val	pending	pending
Test	pending	pending

gradient accumulation. If full fine-tuning remains unstable, the final report will present a diagnostic analysis and use inference-time sampling or frozen-layer adaptation as the backup route. We will not report FID, R-Precision, training loss, or style improvement until those quantities are actually measured.

References

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